

Chapter 2 Homework

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Exercise Group 2.1

- 1) Some roses are red and some violets are blue. *true*
- 2) September has 30 days and there are eight days in a week. *false*
- 3) 3 is less than 4 and 4 is less than 3. *false*
- 4) Napoleon was an emperor of Japan and Marco Polo discovered America. *false*
- 5) $2 + 2 = 5$ and $6 + 3 = 9$. *false*
- 6) $7 + 4 = 11$ and $8 + 0 = 9$. *false*
- 7) $1 + 1 = 2$ and $1 \times 1 = 2$. *false*
- 8) $5 + 1 = 5$ and $5 \times 1 = 6$. *false*

Exercise Group 2.2

- 1) The grass in my yard is not green.
- 2) This rose is not black.
- 3) It is not the case that I am tired.
- 4) The temperature of the sun is less than or equal to 1000 degrees.
- 5) Geometry is not interesting.
- 6) $5 + 4 \neq 9$.
- 7) $6 + 3 \neq 7$.
- 8) $x \neq 2$ or $y \neq 3$.
- 9) $x \neq 2$ and $y \neq 3$.
- 10) Some pigs are not fat.
- 11) All Americans are honest.

Exercise Group 2.3

- 1) Some men in Georgia are less than 6 feet tall.
- 2) Some students in France do not like Geometry.
- 3) Some students in Russia like Geometry.
- 4) Some rectangles are not squares.

- 5) Some triangles are not equilateral.
- 6) Some gadgets are not widgets.
- 7) Some miggles are not maggles.
- 8) There exists some integers a and b s.t $a + b \neq b + a$
- 9) There exists a circle without a center.

Exercise Group 2.4

- 1) No men play golf.
- 2) No boys play football.
- 3) Every triangle is not equilateral.
- 4) For every number x , $x^2 + 2 \neq 5$.
- 5) For every number x , $x + 3 \neq 7$.
- 6) There is no number x such that $x^2 + 2 = 0$.
- 7) All dogs have at least 4 legs.
- 8) There are no negative numbers.
- 9) All freshmen sometimes lie.
- 10) No people work hard.

Exercise Group 2.5

- 1) Today is not Monday or the weather is not cold.
- 2) $x \neq 2$ or $y \neq 5$.
- 3) $2 \times 2 \neq 4$ or $3 \times 5 \neq 15$.
- 4) $\triangle ABC$ is not isosceles or $\triangle ABC$ is not a right triangle.
- 5) The point A is not on the line l or the point B is not on the line m .
- 6) Some cats are not black or some men are not mortal.
- 7) Some cats are not black or all birds are not blue.
- 8) All rectangles are not squares or all triangles are not equilateral.
- 9) All widgets are not gadgets or some wiggles are waggles.

Exercise Group 2.6

- 1) I will not go to the game and I will not go to the movie.
- 2) I will not study hard and I will not fail.
- 3) We will not start at 6:30 and we will not be late.
- 4) A donkey does not have three legs and a rabbit does not have long ears.
- 5) $2 + 2 \neq 5$ and $2 + 2 \neq 6$.
- 6) $3 \times 7 \neq 21$ and $4 \times 5 \neq 20$.

- 7) Billings is wrong and Jones is right.
- 8) None will work and some will not starve.
- 9) Some triangles are not isosceles and some triangles are not right.
- 10) Some roses are red and no violets are blue.

Exercise Group 2.7

- 1) If I eat strawberries, then I will get hives.
- 2) If it rains, then the picnic will be canceled.
- 3) If Jonesevitch plays, we will win the game.
- 4) If a number is larger than 5, then it is positive.
- 5) If $x = y$, then $x^2 = y^2$.
- 6) If $x = 2$ then $3x^2 - 2 = 10$.
- 7) If a geometric figure is a square, then it is not a circle.
- 8) If it's a square, then it's a rectangle.
- 9) If a rectangle has side lengths a and b , then its area is ab .
- 10) If I had a million dollars, then I'd eat steak for breakfast.
- 11) If people don't study mathematics, then they don't have good sense.
- 12) If it rains, then my rheumatism will bother me.
- 13) If I study, then I do well in school.
- 14) If someone doesn't vote, then they are a poor citizen.
- 15) If two lines intersect, then they are not parallel.
- 16) If $3x = 6$, then I know $x = 2$.
- 17) If you are not satisfied, then you get your money back.
- 18) If $x = 2$, then I know $3x = 6$.
- 19) If Jimmy is coming, then I will not go to the party.
- 20) If the shirt shrinks, then your money will be refunded.
- 21) If this shirt will not shrink, then your money will be refunded.
- 22) If we want to remain free, then we must vote in elections.
- 23) If we have rain this week, then the crops will be saved.
- 24) If the crops are saved, then we must have had rain this week.
- 25) If Joe asks me, I will go to the dance.
- 26) If I went to the dance, then Joe asked me.
- 27) If it turns cold, then it will rain tomorrow.
- 28) If it rains tomorrow, then it turned cold.

Exercise Group 2.8

- 1) a) He may or may not be on the team.
b) Fred is not on the team.
- 2) a) He's not a postman.
b) He may or may not be a postman.
- 3) a) Jim is not a sophomore.
b) Joe may be a sophomore.
- 4) a) Jean is a sophomore.
b) George may be given a ticket.
- 5) a) They may have equal bases and they may have equal heights.
b) They have unequal bases.
- 6) a) The house burns down.
b) The house may have burned down.
- 7) a) He may have died in June.
b) Smith may still have died before July 4th.
- 8) Correct
- 9) Incorrect
- 10) Correct
- 11) Correct
- 12) Incorrect. Think "Only humans know algebra."
- 13) Incorrect
- 14) Correct
- 15) Incorrect. It could have been shrunk with a shrink ray.
- 16) Correct
- 17) Correct
- 18) Incorrect. It could have so little honey that it's not sweet.

Exercise Group 2.9

- 1) Fred is not on the team or he must go to bed early.
A man is not a postman or he has foot trouble.
You are not a sophomore or you will be given a ticket.
You will not be given a ticket or you are a sophomore.
Either two triangles do not have equal bases and heights, or they have equal areas.
We will get water or the house will burn down.
A man is not beheaded in June or he will die before July 4th.
- 2) Same answers as given in exercises 8-18 above.
- 3) Fred being on the team is a sufficient condition for Fred going to bed early.
Being a postman is a sufficient condition for having foot trouble.
Being a sophomore is a sufficient condition for receiving a ticket.
Receiving a ticket is a sufficient condition for determining if someone is a sophomore.
Two triangles having equal bases and heights is a sufficient condition for their areas being equal.
Not getting water is a sufficient condition for the house burning down.
Being beheaded in June is a sufficient condition for dying before July 4th.

- 4) Fred going to bed early is a necessary condition for Fred being on the team.
Having foot trouble is a necessary condition for being a postman.
Receiving a ticket is a necessary condition for someone being a sophomore.
Being a sophomore is a necessary condition for receiving a ticket.
Two triangles having equal areas is a necessary condition for them having equal bases and heights.
The house burning down is a necessary condition for us not getting any water.
Dying before July 4th is a necessary condition for being beheaded in June.

Exercise Group 2.10

- 1) If a man is an American citizen, then they were born in Indiana. *false*
- 2) If $x > y$, then $x - y = 3$. *false*
- 3) If two rectangles have equal areas, then they have equal bases and equal heights. *false*
- 4) If $x = 3$ or $x = -3$, then $x^2 = 9$. *true*
- 5) If a man is at least 6 feet tall, then he is a Polynesian. *false*
- 6) If a plant must have sunshine, then it is green. *false*
- 7) If a point is equidistant from the two end points of a line segment, then it must lie on their perpendicular bisector.
true
- 8) If I am breathing, then I am sleeping. *false*
- 9) If $4 + 7 = 10$, then $2 + 4 = 5$. *true* Note: this is vacuously true.
- 10) If $4 + 7 = 11$, then $2 + 4 = 5$. *false*
- 11) If $4 + 7 = 10$, then $2 + 4 = 6$. *true*
- 12) If $4 + 7 = 11$, then $2 + 4 = 6$. *true*

Exercise Group 2.11

- 1) Equivalent
- 2) Not equivalent
- 3) Equivalent
- 4) Equivalent
- 5) Not equivalent
- 6) Equivalent
- 7) Equivalent
- 8) Point P lying on the circle with center at O and a radius of five inches is a necessary and sufficient condition for segment OP having a length of five inches.
Line l being parallel to line m is a necessary and sufficient condition for line n being parallel to line m .
 $x > y$ is a necessary and sufficient condition for $y < x$.
 $2x + 3 = 17$ is a necessary and sufficient condition for $x = 7$.
A geometric figure being square is a necessary and sufficient condition for a geometric figure being a special parallelogram.
 $\triangle ABC$ being equilateral is a necessary and sufficient condition for $\triangle ABC$ being equiangular.
 $x = 3$ or $x = -3$ is a necessary and sufficient condition for $x^2 = 9$.

Exercise Group 2.12

- 1) $x = 5$ and $y = 2$.
- 2) $y < x$.
- 3) Line segments AB and CD have no points in common.
- 4) Lines l and m intersect.
- 5) You will like Dum-Dums.
- 6) This figure is a parallelogram.
- 7) That polygon has 3 sides.
- 8) AB is a line segment.
- 9) The segment AB does not intersect line l .
- 10) The segment AB intersects l . (I think this problem meant to say A and B lie on opposite sides of l)

Exercise Group 2.13

- 1) If oranges are not composed of uranium, then $2 \times 2 \neq 4$.
- 2) If I won't eat my hat, then you can not run a mile in 4 minutes.
- 3) If we will not have a cold spring, then the ground hog does not see his shadow.
- 4) If there is life on Mars, then there is an atmosphere on Mars.
- 5) If P is on line s , then it is not on line r .
- 6) If your money is not returned, then you are satisfied.
- 7) If food does not have honey in it, then it is not sweet.
- 8) If you are given a ticket, then you don't always obey traffic laws.
- 9) There being an atmosphere on Mars is a necessary condition for there being life.
- 10) There being life on Mars is a sufficient condition for there being an atmosphere.

Exercise Group 2.14

- 1) We will stay home if and only if the weather is bad.
- 2) You will get a ticket if and only if you are a sophomore.
- 3) We will not stay home if and only if the weather is not bad.
- 4) You will pass the course if and only if you receive more than 50% on this test.
- 5) Your dog will obey you if and only if you treat him well.
- 6) I will marry you if and only if you buy me a mink coat.
- 7) Food is sweet if and only if it has honey in it.

Exercise Group 2.15

1) *Proof.* Uncle George will take us skating today.

1. Mother says we may go $\rightarrow$ uncle George will take us skating today (hypothesis 1).
2. Mother says we may go (hypothesis 2).
3. Therefore, uncle George will take us skating today (combine statements 1 and 2). ☐

2) *Proof.* There is no animal life on the moon.

1. There is no oxygen on the moon $\rightarrow$ there is no animal life on the moon (hypothesis 1).
2. There is no oxygen on the moon (hypothesis 2).
3. Therefore, there is no animal life on the moon (combine statements 1 and 2). ☐

3) Strictly speaking, there is not enough information. Joan may not be a person and could be the name of my cat.

Proof. Joan is a person who can not be trusted.

1. Anyone being a person with green eyes $\rightarrow$ they are a person who can not be trusted (hypothesis 1).
2. Joan being a person with green eyes $\rightarrow$ Joan is a person who can not be trusted (statement 1).
3. Joan is a person with green eyes (hypothesis 2).
4. Therefore, Joan is a person who can not be trusted (combine statements 2 and 3). ☐

Note: we will assume for the remainder of the problems that all proper names refer to people.

4) *Proof.* Henry does not have blue eyes.

1. A person has blue eyes $\rightarrow$ they are honest (hypothesis 1).
2. A person is dishonest $\rightarrow$ they don't have blue eyes (contrapositive of statement 1).
3. Henry is dishonest $\rightarrow$ Henry does not have blue eyes (statement 2).
4. Henry is dishonest (hypothesis 2).
5. Therefore, Henry does not have blue eyes (combine statements 3 and 4). ☐

5) Not enough information. It could still be the case that Joe parts his hair on the right.

6) *Proof.* Simba does not have green whiskers.

1. Simba has green whiskers $\rightarrow$ Simba has a long tail (hypothesis 1).
2. Simba does not have a long tail $\rightarrow$ Simba does not have green whiskers (contrapositive of statement 1).
3. Simba does not like milk $\rightarrow$ Simba does not have a long tail (hypothesis 2).
4. Simba does not like milk (hypothesis 3).
5. Simba does not have a long tail (combine statements 3 and 4).
6. Therefore, Simba does not have green whiskers (combine statements 5 and 2). ☐

7) *Proof.* Kate did not attend MIT last year.

1. Kate attended MIT last year $\rightarrow$ Kate spent more than \$12,000 last year (hypothesis 1).
2. Kate spent less than or equal to \$12,000 last year $\rightarrow$ Kate did not attend MIT last year (contrapositive of 1).
3. Kate spent \$980 last year $\rightarrow$ Kate spent less than or equal to \$12,000 last year (arithmetic).
4. Kate spent \$980 last year (hypothesis 2).
5. Kate spent less than or equal to \$12,000 last year (combine 3 and 4).
6. Therefore, Kate did not attend MIT last year (combine 5 and 2). ☐

8) *Proof.* $x \neq 11$.

1. $x - y = 4$ (hypothesis 2).
2. $x = y + 4$ (arithmetic on 1).
3. $x + y = 20$ (hypothesis 1).
4. $(y + 4) + y = 20$ (substitute 2 into 3).
5. $(y + 4) + y = 20 \leftrightarrow 2y + 4 = 20 \leftrightarrow 2y = 16 \leftrightarrow y = 8$ (arithmetic on 4).
6. $x = 8 + 4$ (substitute 5 into 2)
7. Therefore, $x = 12 \neq 11$, and so $x \neq 11$ (arithmetic on 6) ☐

9) *Proof.* $3x - y = 10$.

1. Observe that $x + 2y = 3 \rightarrow x = -2y + 3$ (hypothesis 2).

2. Observe that $2x - 3y = 7$ (hypothesis 1).

3. Then $2x - 3y = 7 \rightarrow 2(-2y + 3) - 3y = 7 \rightarrow -4y + 6 - 3y = 7 \rightarrow -7y + 6 = 7 \rightarrow -7y = 1 \rightarrow y = -\frac{1}{7}$ (substitute 1 into 2).

4. Then $x = -2y + 3 \rightarrow x = -2(-\frac{1}{7}) + 3 \rightarrow x = \frac{2}{7} + 3 \rightarrow x = \frac{2}{7} + \frac{21}{7} \rightarrow x = \frac{23}{7}$ (substitute 3 into 1).

5. Then $3x - y = 3\frac{23}{7} - (-\frac{1}{7}) = \frac{69}{7} + \frac{1}{7} = \frac{70}{7} = 10$ (substitute in x and y).

6. Therefore $3x - y = 10$. □

10) *Proof.* Joe does not have more than four yellow marbles left.

1. Let y be the number of yellow marbles and let b be the number of blue marbles Joe has after he gave his sister 6.

2. Observe that $y + b = 11 - 6 \rightarrow y + b = 5$ (given).

3. Observe that $b \geq 1$ (given).

4. Then from line 1 we get that $b = 5 - y$ and from 2 we get $b \geq 1$.

5. Then, substituting for b we get $5 - y \geq 1 \rightarrow 5 \geq 1 + y \rightarrow 4 \geq y$.

6. Therefore, Joe does not have more than four yellow marbles left. □

11) There is not enough information to prove Sally will get an A, all we know is that Jim will not.

Proof. Jim will not receive an A.

1. Getting an A $\rightarrow$ getting an average score of more than 90 (given).

2. Getting an average score of 90 or less $\rightarrow$ not getting an A (contrapositive of 1).

2. Jim's test grades have been 89, 84, 92, 85, and 100 (given).

3. Jim's average score is $(89 + 84 + 92 + 85 + 100)/5 = 90 \leq 90$, and so Jim got an average score of 90 or less (line 2, and definition of average).

4. Therefore, Jim will not get an A (combine 2 and 3). □

12) There's not enough information. All we know is that if Amanda marries Silas, then he bought her the requested items. We do not know if Amanda married him.

13) *Proof.* $y = 4$.

1. Observe that x and y are positive whole numbers such that $x + y = 10$ is given.

2. Then all the possible values for (x, y) are (9,1), (8,2), (7,3), (6,4), (5,5), (4,6), (3,7), (2,8), and (1,9).

3. Since it's given that $x > y$, then the possible values are now restricted to (9,1), (8,2), (7,3), and (6,4).

4. Since it's given that $y + 4 > x$, then the only possible value is now (6,4).

5. Therefore $x = 6$ and $y = 4$. □

14) *Proof.* There are numbers which are not positive or negative or equal to zero.

1. Observe that k is a number and that $k^2 = -1$ (given).

2. k is positive $\rightarrow k^2 \neq -1$ (given).

3. $k^2 = -1 \rightarrow k$ is not positive (contrapositive on 2).

4. k is negative $\rightarrow k^2 \neq -1$ (given).

5. $k^2 = -1 \rightarrow k$ is not negative (contrapositive on 4).

6. $k = 0 \rightarrow k^2 \neq -1$ (given).

7. $k^2 = -1 \rightarrow k \neq 0$ (contrapositive on 6).

8. Then, since $k^2 = -1$, we have that k is a number and k is not positive and k is not negative and $k \neq 0$.

9. Therefore, there are numbers which are not positive or negative or equal to zero. □

15) Not enough information; b could be 0.

Chapter 2 Review Exercises

- 1) $\alpha \wedge \beta$ is *true* iff both α and β are *true*, otherwise it is *false*.
- 2) If you must pass Calculus 1 or pass Physics 1 to take my course, then it is ok if you passed both Calculus 1 and Physics 1.
- 3) The negation of not α is just α . In other words $\neg\neg\alpha \longleftrightarrow \alpha$.
- 4) "I am the teacher and I will fail you." to "Either I'm not the teacher, or I will not fail you".
- 5) "You are going to pass the course or I am going to throw a marker at you." to "You are going to fail the course and I will not throw a marker at you."
- 6) $\alpha \rightarrow \beta$ means that if α is *true* then β also has to be *true*.
"If $x = 2$ then $x^2 = 4$." (*true*) and "If $x^2 = 4$ then $x = 2$." (*false*)
- 7) Since α is a statement, its only possible values are *true* or *false*. If $\alpha = \text{true}$ then

$$\alpha \vee \neg\alpha \leftrightarrow \text{true} \vee \neg\text{true} \leftrightarrow \text{true} \vee \text{false} \leftrightarrow \text{true}$$

and if $\alpha = \text{false}$ then

$$\alpha \vee \neg\alpha \leftrightarrow \text{false} \vee \neg\text{false} \leftrightarrow \text{false} \vee \text{true} \leftrightarrow \text{true},$$

and so in either case, it is *true*. Example: "The sky is blue or the sky is not blue" is always true.

- 8) *Proof.* When $(\alpha \wedge \beta) \vee \gamma = \text{T}$

α	β	γ	$\alpha \wedge \beta$	$(\alpha \wedge \beta) \vee \gamma$
T	T	T	T	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	T
F	F	F	F	F

□

Example: "The sky is blue and the grass is purple, or $2 + 2 = 4$."

- 9) If $x = 2$ then $2x = 4$. (the converse is also true)
If $x = 2$ then $x^2 = 4$. (the converse is not true)
- 10) (a) If a man is rich, then he is from Chicago.
(b) If I go to church, then she will sing in the choir.
(c) If Smith got ulcers, then he was elected president.
(d) If you're a sophomore, then you are taking geometry.
(e) If you're brave, then you're free.
- 11) (a) If a man is not from Chicago, then he is not rich.
(b) If she will not sing in the choir, then I will not go to church.
(c) If Smith was not elected president, then he did not get ulcers.
(d) If you're not taking Geometry, then you are not a sophomore.
(e) If you're not free, then you're not brave.
- 12) (a) If food is sweet, then it has honey in it.
(b) If the weather is warm, then my rheumatism does not bother me.
(c) If you are healthy, then you drink milk.

- (d) If you got a grade of 80% on your final exam, then you passed the course.
- (e) If you are at the dance, then you are a sophomore.
- 13) (a) Honesty is not the best policy.
- (b) Some math teachers are not brutes.
- (c) The potato crop will not fail or some people will not starve.
- (d) Either roses are not red, violets are not blue, sugar is not sweet, or I do not love you.
- (e) No cats can swim, or some chickens can.
- 14) $\alpha \wedge \beta$
- 15) $\alpha \rightarrow \beta$
- 16) $\beta \rightarrow \alpha$
- 17) $\alpha \leftrightarrow \beta$
- 18) $\beta \rightarrow \alpha$
- 19) $\beta \rightarrow \alpha$
- 20) $\alpha \vee \beta$
- 21) $\alpha \wedge \beta$

Chapter 2 Algebra Review

1) *Proof.* If $x = 7$ then $4x + 5 = 33$.

1. Let $x = 7$.

2. Then $4x + 5 = 4(7) + 5 = 28 + 5 = 33$. □

2) *Proof.* If $4x + 5 = 33$ then $x = 7$.

1. Let $4x + 5 = 33$.

2. Then $4x + 5 = 28 + 5$, and by using property 1 we get $4x = 28$.

3. Then $4x = 4(7)$, and by using property 2 we get $x = 7$. □

3) Prove that if $7x + 9 = 65$ then $x = 8$, and conversely.

Proof. If $7x + 9 = 65$ then $x = 8$.

1. Let $7x + 9 = 65$.

2. Then $7x + 9 = 56 + 9$, and by using property 1 we get $7x = 56$.

3. Then $7x = 7(8)$, and by using property 2 we get $x = 8$. □

Proof. If $x = 8$ then $7x + 9 = 65$.

1. Let $x = 8$.

2. Then $7x + 9 = 7(8) + 9 = 56 + 9 = 65$. □

4) Prove that $5x + (-3) = 27$ only if $x = 6$.

Proof. If $5x + (-3) = 27$ then $x = 6$.

1. Let $5x + (-3) = 27$.

2. Then $5x + (-3) = 30 + (-3)$, and by using property 1 we get $5x = 30$.

3. Then $5x = 5(6)$, and by using property 2 we get $x = 6$. □

5) Prove that $x = 11$ is a necessary condition for $5x + 4 = 59$.

Proof. If $5x + 4 = 59$ then $x = 11$.

1. Let $5x + 4 = 59$.

2. Then $5x + 4 = 55 + 4$, and by using property 1 we get $5x = 55$.

3. Then $5x = 5(11)$, and by using property 2 we get $x = 11$. □

6) *Proof.* If $x = 11$ then $5x + 4 = 59$.

1. Let $x = 11$.

2. Then $5x + 4 = 5(11) + 4 = 55 + 4 = 59$. □

7) *Proof.* Prove that $2x + 7 = 19$ iff $4x - 3 = 21$.

1. ($\Rightarrow$) Let $2x + 7 = 19$.

2. Then $2x + 7 = 12 + 7$, and by using property 1 we get $2x = 12$.

3. Then $2x = 2(6)$, and by using property 2 we get $x = 6$.

4. Then $4x - 3 = 4(6) - 3 = 24 - 3 = 21$.

5. ($\Leftarrow$) Let $4x - 3 = 21$, or $4x + (-3) = 21$.

6. Then $4x + (-3) = 24 + (-3)$, and by using property 1 we get $4x = 24$.

7. Then $4x = 4(6)$, and by using property 2 we get $x = 6$.

8. Then $2x + 7 = 2(6) + 7 = 12 + 7 = 19$. □

- 8) Prove that there is no number x such that $3x + 5 = 11$ and $4x - 1 = 3$.

Proof. If x is a number then either $3x + 5 \neq 11$ or $4x - 1 \neq 3$.

1. Let x be a number, and observe that if $3x + 5 \neq 11$ then we are done.
2. Assume $3x + 5 = 11$.
3. Then $3x + 5 = 6 + 5$, and by property 1 we get $3x = 6$.
4. Then $3x = 3(2)$, and by property 2 we get $x = 2$.
5. Then $4x - 1 = 4(2) - 1 = 8 - 1 = 7$, and so $4x - 1 \neq 3$.
6. Therefore, if x is a number then either $3x + 5 \neq 11$ or $4x - 1 \neq 3$. □

- 9) What can you infer if it is known that $a \neq 0$ and $a(3b - 8) = a$?

Proof. If $a \neq 0$ and $a(3b - 8) = a$, then $b = 3$.

1. Assume $a \neq 0$ and $a(3b - 8) = a$.
2. then $a(3b - 8) = a(1)$, and by property 2 we get $3b - 8 = 1$.
3. Then $3b + (-8) = 9 + (-8)$, and by using property 1 we get $3b = 9$.
4. Then $3b = 3(3)$, and by using property 2 we get $b = 3$. □

- 10) What can you prove if it is known that $r(7b - 5) = r$? Note: the author intended for us to think about r and b as integers.

Proof. If $r(7b - 5) = r$ then $r = 0$.

1. We will show the contrapositive; if $r \neq 0$ then $r(7b - 5) \neq r$.
2. Assume that $r \neq 0$, and we will show that it's impossible to have $r(7b - 5) = r$. In other words, we will show that the equation $r(7b - 5) = r$ implies a contradiction and so must not be true.
3. If $r(7b - 5) = r$, then by property 2 we get $7b - 5 = 1$.
4. Then $7b - 5 = 6 - 5$, and so by property 1 we get $7b = 6$.
5. Then $7b = 7(\frac{6}{7})$, and so by property 2 we get $b = \frac{6}{7}$.
6. This contradicts the fact that b is an integer.
7. So, if $r \neq 0$ then $r(7b - 5) \neq r$.
8. Therefore, if $r(7b - 5) = r$ then $r = 0$. □

- 11) It is given that $(3a + 12)(2b - 8) = 0$ and $b \neq 4$. What can be proved?

Proof. If $(3a + 12)(2b - 8) = 0$ and $b \neq 4$, then $a = -4$.

1. Assume that $(3a + 12)(2b - 8) = 0$ and $b \neq 4$.
2. Observe that if $2b - 8 = 0$, then $2b - 8 = 8 - 8 \rightarrow 2b = 8 \rightarrow 2b = 2(4) \rightarrow b = 4$.
3. Since $b \neq 4$, then by the contrapositive $2b - 8 \neq 0$.
4. Then $(3a + 12)(2b - 8) = 0 \rightarrow (3a + 12)(2b - 8) = (2b - 8) \cdot 0$.
5. Then by property 2 we get $3a + 12 = 0$, which gives us $3a + 12 = (-12) + 12$.
6. Then by property 1 we get $3a = -12$, which gives us $3a = 3(-4)$.
7. Then by property 2 we get $a = -4$.
8. Therefore, if $(3a + 12)(2b - 8) = 0$ and $b \neq 4$, then $a = -4$. □

- 12) It is known that $(2a - 3)(b - 3) = 2a - 2$.

Proof. If $a = 1$ then $b = 3$.

1. Assume that $a = 1$.
2. Then $(2 \cdot 1 - 3)(b - 3) = 2 \cdot 1 - 2 \rightarrow (2 - 3)(b - 3) = 2 - 2 \rightarrow (-1)(b - 3) = 0 \rightarrow (-1)(b - 3) = (-1)0$.
3. Then by property 2 we get $b - 3 = 0 \rightarrow b + (-3) = 3 + (-3)$.
4. Then by property 1 we get $b = 3$.
5. Therefore, if $a = 1$ then $b = 3$. □

Proof. If $a \neq 1$ then $b \neq 3$.

1. We will prove the contrapositive; if $b = 3$ then $a = 1$.
2. Assume that $b = 3$.
3. Then $(2a - 3)(3 - 3) = 2a - 2 \longrightarrow (2a - 3)0 = 2a - 2 \longrightarrow 0 = 2a - 2 \longrightarrow 2 + (-2) = 2a + (-2)$.
4. Then by property 1 we get $2 = 2a \longrightarrow 2(1) = 2a$.
5. Then by property 2 we get $1 = a$, or $a = 1$.
6. Therefore, if $a \neq 1$ then $b \neq 3$. □

The following proof is beyond the scope of this chapter and relies upon common algebraic properties. You were only required to discover that if $a \neq 1$, then $a = 2$ and $b = 5$ is the solution.

Proof. If $a \neq 1$ then $a = 2$ and $b = 5$.

1. Assume that $a \neq 1$, and so $2a - 2 \neq 0$.
2. Since a is an integer, we also know that $2a - 3 \neq 0$ (otherwise we get $a = \frac{3}{2}$)
3. Then $(2a - 3)(b - 3) = 2a - 2 \iff (2a - 3)(b - 3) - 1 = 2a - 3 \iff \frac{(2a-3)(b-3)-1}{2a-3} = 1$.
4. Then $(b - 3) - \frac{1}{2a-3} = 1$.
5. Observe that $\frac{1}{2a-3}$ must be an integer since $b - 3$ is an integer and 1 is an integer.
6. The only way for $\frac{1}{2a-3}$ to be an integer is for its denominator to be 1, so we know $2a - 3 = 1$.
7. Then $2a - 3 = 1 \longrightarrow 2a = 4 \longrightarrow a = 2$.
8. Substituting 2 for a in the initial equation gives us $(2 \cdot 2 - 3)(b - 3) = 2 \cdot 2 - 2$.
9. Then $(4 - 3)(b - 3) = 4 - 2 \longrightarrow (1)(b - 3) = 2 \longrightarrow b - 3 = 2 \longrightarrow b = 5$.
10. Therefore, if $a \neq 1$ then $a = 2$ and $b = 5$. □